Normalized Transfer of a Nonlinear Dispersion Tail

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Abstract

We give the exact event-inclusion and normalized-measure theorem transporting the dependent Gaussian dispersion tail to a convex nonlinear $SU(2)$ chart.

Let G be the local cylinder, let D be nonlinear Fréchet dispersion, and let Q be the centered linear dispersion. Suppose each staple logarithm has Taylor remainder at most ϵ, the metric factor is s_r, and

$$\mu(A \cap G) \le e^{\ell}\nu(A \cap G), \qquad \mu(G^c) \le q_{\mathrm{exc}}.$$

Theorem 1. *For m staples and $\eta > 0$,*

$$\{D \ge m\eta^2\} \cap G \subseteq \{Q \ge m(s_r\eta - \epsilon)_+^2\},$$

and therefore

$$\mu\{D \ge m\eta^2\} \le e^{\ell}\nu\{Q \ge m(s_r\eta - \epsilon)_+^2\} + q_{\mathrm{exc}}.$$

Proof. Stage 41 gives $P \ge s_r^2 D$. Centering is a contraction, so the staple Taylor remainder gives $\sqrt{P} \le \sqrt{Q} + \sqrt{m}\epsilon$. Rearrangement proves the inclusion. Split the event over G and G^c and apply the two normalized premises. □

At nonlinear RMS threshold 0.006, the frozen centered reference meets the allocated high-channel budget after chart loss.

Remark 1. *The density and complement premises have not been proved uniformly over physical retained boundary states; the numerical closure is restricted.*